# MILITARY SPECIFICATION

## BEAKER, LABORATORY, Conical, Polystyrene, Disposable

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a disposable 2-milliliter (ml.) polystyrene beaker which has a flat bottom and a conical interior.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-636 | Boxes, Shipping, Fiberboard. |
| PPP-B-676 | Boxes, Setup. |

Military

| | |
|---|---|
| MIL-P-116 | Preservation-packaging, Methods of. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

FSC 6640

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D703 Polystyrene Molding and Extrusion Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Beaker shall be fabricated of clear polystyrene in accordance with ASTM D703.

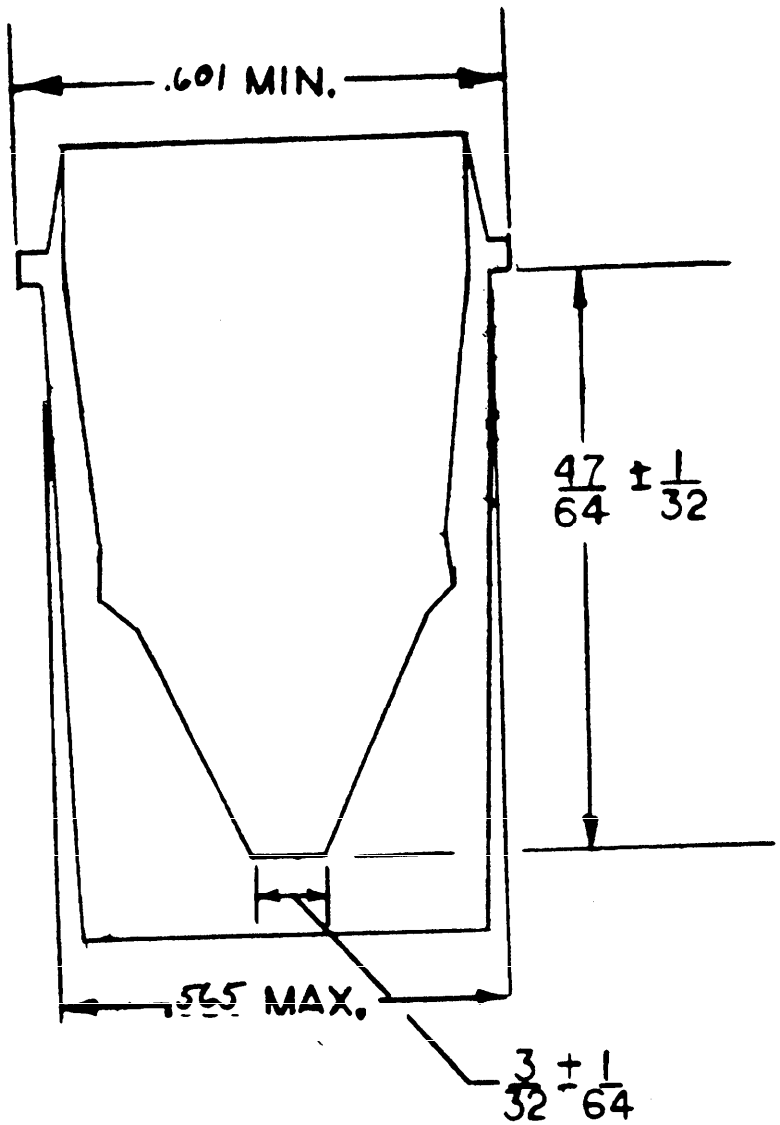
3.2 Construction. Design shall be as shown in Figure 1. Nominal outside diameter of flange shall be 5/8 inch (in.) (16 millimeters) (mm.); nominal outside diameter directly below the flange shall be 9/16 in. (14 mm.). Exact dimensions shall be as shown in Figure 1.

3.3 Capacity and graduation markings. Beaker shall be marked at 0.5, 1.0 and 1.5 ml. increments. Junction of internal cone with outside wall may be used for the 0.5 ml. mark. When tested as specified in 4.4.1, full beaker shall contain 2.0 ml. $\pm$ .2 ml. In addition, beaker shall contain the applicable amount, $\pm$ 10%, for each of the graduation marks.

3.4 Workmanship. Workmanship shall be first class throughout. Beaker shall be free of defects listed in Table III and any other defects which detract from their appearance or may impair their serviceability.



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DIMENSIONS IN INCHES

FIGURE 1. BEAKER, LABORATORY, CONICAL, POLYSTYRENE.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. Unit of product shall be the beaker.

Table I Sampling for examination

| | Inspection level | AQL (percent defective) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 2.5 |
| Minor defects | II | 4.0 |
| For dimensional examination | S-1 | 2.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be the beaker, and acceptance number shall be zero.



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Table II Sampling for tests

| Characteristic | Requirement | Test procedure | Inspection level |
|---|---|---|---|
| Accuracy | 3.3 | 4.4.1 | S-4 |

4.3 Examination. The beaker shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with, but shall not be limited to, the defects listed in Table III.

Table III Classification of defects

| Category | Defect |
|---|---|
| Major | |
| 101 | Beaker not free of wobble when placed on a smooth, flat surface. |
| 102 | Beaker not free of deformation. |
| 103 | Beaker not free of cracks and tears. |
| 104 | Beaker not free of sharp and feathered edges. |
| 105 | Beaker not free of crazing. |
| Minor | |
| 201 | Beaker not free of protrusions. |
| 202 | Beaker not free of dirt and foreign matter. |

4.3.1 Dimensional examination. Beakers shall be examined for defects in dimensions. Any dimensions not within the tolerance specified shall be considered a defect.

4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The total capacity and accuracy of all graduated increments shall be tested by gravimetric or volumetric methods.

5. PACKAGING.

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified (see 6.2).



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5.1.1 Unit of issue. One package containing one thousand beakers, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Beakers shall be cleaned in accordance with MIL-P-116, method C-1.

5.1.2.2 Drying. Beakers shall be dried in accordance with MIL-P-116.

5.1.2.3 Unit packaging. One thousand beakers shall be neatly packaged in accordance with MIL-P-116, method III in a clear snug-fitting, sealed polyethylene bag having a minimum wall thickness of 2.0 mil. At the option of the contractor, the bag may be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Beakers shall be clean, dry and packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Ten unit packages (10,000 beakers) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Ten unit packages (10,000 beakers) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged beakers shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be



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shown. When the polyethylene bag is the unit package, markings shall be printed on the bag or on a label placed within the bag in such a manner so as to be readily identified.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Beaker is intended for use as a sample container for automatic analyzers and for other laboratory procedures.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item listed in the Federal Supply Catalog:

| National Stock No. | Item Identification |
|---|---|
| 6640-01-009-3762 | BEAKER, LABORATORY, Conical, Polystyrene, Disposable, 2 ml., 1000s |

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6640-0966



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-B-37405

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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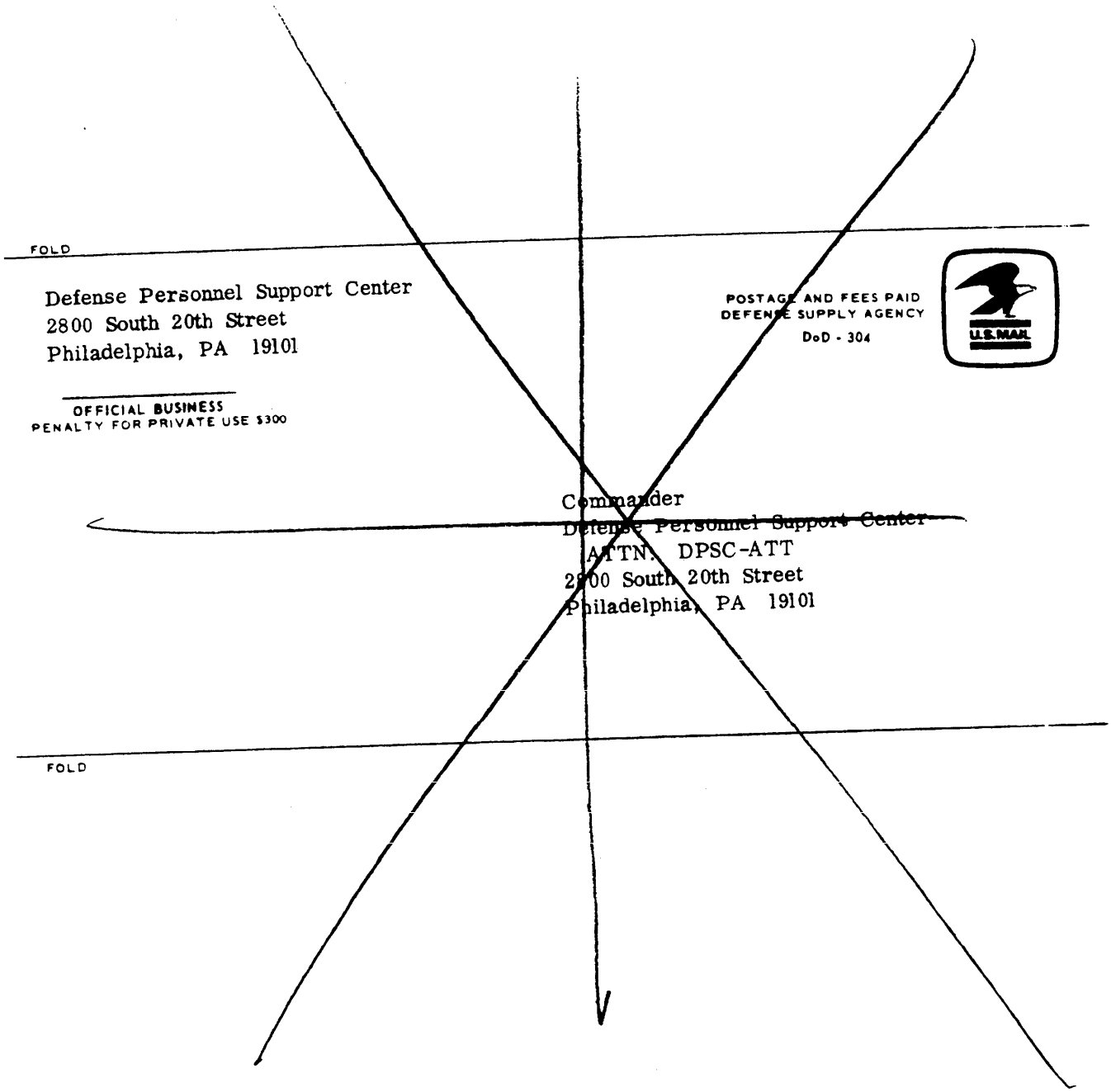
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